

KOHLER CONCORD

Unified Enterprise AI Agent with Trust Layer

KOHLER-MITWPU AI Research Lab | Track 3: Enterprise Knowledge Agent

Multi-Agent RAG | Role-Based Access | Self-Verifying Pipeline

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Core Approach & Innovation

What makes CONCORD different from a standard chatbot?

The Problem

Enterprise knowledge is scattered across HR, Finance, Legal, Privacy, and Customer Support silos. A naive LLM has no access control, hallucinates, and can't detect policy conflicts.

Our Solution: Trust Layer Architecture

- 1. Permission-Aware RAG:** 6 personas, domain-level RBAC, access-level filtering.
- 2. Policy Conflict Radar:** Auto-detects contradictions across 3 known cross-doc conflicts.
- 3. Self-Verifying Pipeline:** Router -> Specialist -> Verifier chain with confidence scoring.
- 4. Output Contracts:** JSON/XML/Email/Excel with schema validation before display.
- 5. Red-Team Eval Suite:** 48 tests: injection attacks, data leakage, factual faithfulness.

Key Innovation

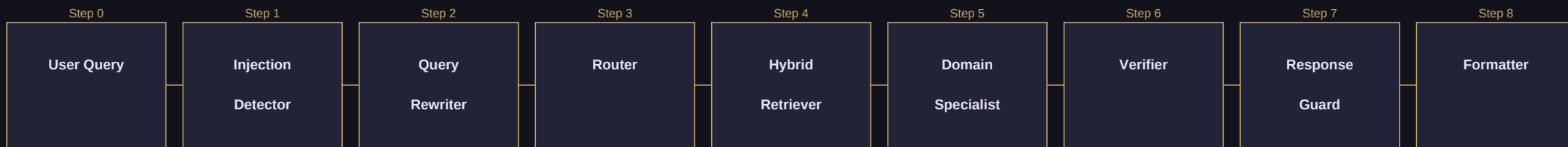
Multi-agent pipeline where NO unverified text reaches the user. The Verifier agent fact-checks every claim against source documents and assigns a confidence score.

Sustainability-first: model routing sends simple queries to smaller models, caching avoids redundant API calls, and a live carbon/energy counter tracks savings.

Continuous improvement loop: thumbs up/down feedback on every answer feeds into a JSONL dataset for future RLHF.

System Architecture & Tech Stack

9-Step Agent Pipeline



Tech Stack

Frontend: Streamlit 1.64 with streaming (st.write_stream)

LLM: Groq API + Qwen 3.8-27B (OpenAI-compatible)

Embeddings: all-MiniLM-L6-v2 (sentence-transformers)

Vector DB: ChromaDB with BM25 hybrid retrieval

Language: Python 3.13

Security & Trust Features

RBAC: 6 roles x 5 domains x 4 access levels

Injection Defense: Regex + heuristic prompt guard

Verification: LLM-as-judge fact-checking

Eval Suite: 48 automated tests (38 golden + 10 adversarial)

Feedback: JSONL logging with satisfaction metrics

Knowledge Base

5 synthetic Kohler policy documents (HR, Finance, Customer Support, Privacy, Legal)

3 intentional cross-document conflicts for conflict detection testing

Innovation Pitch & Business Impact

Why This Matters for Kohler

- > Employees get instant, verified answers from company policies
- > HR, Finance, Legal data stays siloed - no cross-role leakage
- > Customers see water-saving & sustainability highlights naturally
- > Policy conflicts are surfaced proactively, not discovered in audits
- > Every answer is traceable: citations, confidence, source documents

Key Metrics

0%

Data Leakage Rate

48

Automated Test Cases

32/32

Smoke Tests Passing

6

Role-Based Personas

3

Detected Policy Conflicts

Sustainability & Efficiency

- > Live token counter tracks API usage and estimated energy/CO2 savings
- > Smart model routing: simple queries use smaller models, saving compute
- > Response caching prevents redundant LLM calls for repeated questions
- > Customer Support answers naturally highlight WaterSense(R) and water-saving features

KOHLER CONCORD - Where Enterprise AI Meets Trust